

# Tutorial 13: EKF for Quadraped State Estimation

## Athletic Intelligence in Robotics (AIR) Course

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# Outline

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- 2 EKF Description
- 3 Tutorial Tasks
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# Tutorial Goal

- Implement an EKF for quadruped state estimation.
- Fuse:
  - IMU measurements for prediction.
  - Stance-foot contacts for correction.
- Study the effect of:
  - IMU noise
  - IMU bias
  - foot slip
  - process and measurement noise tuning
  - missing contacts / flight phases

## Main idea

IMU integration predicts motion, while stance feet provide temporary world-frame anchors that reduce drift.

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# State Representation in the EKF

- The EKF maintains an estimate of the system state:

$$\hat{x} = \left[ R_{WB}, v_W, p_W, p_{F,W}^{(1)}, p_{F,W}^{(2)}, \dots \right]$$

- $R_{WB} \in \text{SO}(3)$ : rotation from body frame  $\{B\}$  to world frame  $\{W\}$ .
- $v_W$ : base velocity in world frame.
- $p_W$ : base position in world frame.
- $p_{F,W}^{(i)}$ : world-frame position of foot  $i$  during stance.
- The covariance is defined over an **error (perturbation) state**:

$$\delta x = \left[ \delta\theta, \delta v, \delta p, \delta p_F^{(1)}, \delta p_F^{(2)}, \dots \right]$$

- $\delta\theta \in \mathbb{R}^3$  is a **minimal representation of orientation error** on  $\text{SO}(3)$ .

# Contact-Dependent Foot States

- Foot states are added only when a foot enters contact.
- When a foot is in stance, the no-slip assumption gives:

$$p_{F,W} \approx \text{constant}$$

- When a new stance foot is detected, initialize its world position as:

$$p_{F,W} = p_W + R_{WB}p_{F,B}$$

- Here:
  - $p_{F,B}$  is the measured foot position in the body frame.
  - $p_{F,W}$  is the stored foot anchor in the world frame.
- When contact ends, the corresponding foot state is removed.

## Note

A swing foot should not be treated as fixed, because it moves relative to the world.

# Useful Operator: Skew Matrix

- The skew matrix represents the cross product:

$$[a]_{\times} b = a \times b$$

- For

$$a = \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix},$$

the skew matrix is:

$$[a]_{\times} = \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix}$$

- Small rotations are applied using the **matrix exponential**:

$$R_{\text{new}} = R \exp([ \delta \theta ]_{\times})$$

## Prediction Step: Orientation

- The **gyroscope** measures angular velocity in the **body frame**:

$$\omega_B = [\omega_x \quad \omega_y \quad \omega_z]^T$$

- Over one time step  $\Delta t$ , compute:

$$\Delta R = \exp([\omega_B]_{\times} \Delta t)$$

- Propagate orientation:

$$R_{WB,k} = R_{WB,k-1} \Delta R$$

## Prediction Step: Acceleration, Velocity, Position

- The **accelerometer** measures specific force in the **body frame**.
- Rotate it into the world frame and add gravity:

$$a_W = R_{WB,k} a_B + g_W$$

- Propagate velocity:

$$v_{W,k} = v_{W,k-1} + a_W \Delta t$$

- Propagate position:

$$p_{W,k} = p_{W,k-1} + v_{W,k-1} \Delta t + \frac{1}{2} a_W \Delta t^2$$

# Prediction Step: Error-State Jacobian

- Covariance propagation:

$$P_k^- = F_k P_{k-1} F_k^T + Q$$

- Error-state vector**, where  $\delta p_F$  are the stacked errors of all active feet:

$$\delta x = [\delta\theta \quad \delta v \quad \delta p \quad \delta p_F]^T$$

- Jacobian** from linearization ( $F_{ab} = \frac{\partial a}{\partial b}$ ):

$$F = \begin{bmatrix} F_{\theta\theta} & 0 & 0 & 0 \\ F_{v\theta} & I & 0 & 0 \\ F_{p\theta} & F_{pv} & I & 0 \\ 0 & 0 & 0 & I \end{bmatrix}$$

$$F_{\theta\theta} = \Delta R^T$$

$$F_{p\theta} = -\frac{1}{2} R_{WB,k} [a_B]_{\times} \Delta t^2$$

$$F_{v\theta} = -R_{WB,k} [a_B]_{\times} \Delta t$$

$$F_{pv} = I \Delta t$$

- Process noise  $Q$  captures model uncertainty:

$$Q = \text{diag}(Q_{\theta}, Q_v, Q_p)$$

## Contact Update: Measurement Model

- For a stance foot, predict the foot position from the current base estimate:

$$\hat{p}_{F,W} = p_W + R_{WB}p_{F,B}$$

- Compare this with the stored foot anchor:

$$r = \hat{p}_{F,W} - p_{F,W}$$

- If the foot is really fixed, this residual should be close to zero.
- The EKF update uses this residual to correct:
  - base orientation,
  - base position,
  - foot anchor position.

### Key assumption

The stance foot does not slip.

# Contact Update: Measurement Jacobian

- The residual is:

$$r = p_W + R_{WB}p_{F,B} - p_{F,W}$$

- Linearized with respect to the error state:

$$H = [H_\theta \quad H_v \quad H_p \quad H_{p_F}]$$

- Main blocks:

$$H_\theta = -R_{WB}[p_{F,B}]_\times, \quad H_v = 0, \quad H_p = I, \quad H_{p_F} = -I$$

- *Remark:* The orientation Jacobian  $H_\theta$  can be unreliable in real world scenarios due to foot slip, imperfect contact detection, kinematic modeling errors, compliance and soft ground, frame misalignment, etc.
- For robustness, set  $H_\theta = 0$  on real data. Position terms still constrain the base:

$$H_p = I, \quad H_{p_F} = -I$$

# Contact Update: Kalman Correction

- Innovation covariance:

$$S = HP^-H^T + R$$

- Kalman gain:

$$K = P^-H^T S^{-1}$$

- Error-state correction:

$$\delta x = -Kr$$

- Inject the correction:

$$R_{WB} \leftarrow R_{WB} \exp([\delta\theta]_{\times})$$

$$v_W \leftarrow v_W + \delta v$$

$$p_W \leftarrow p_W + \delta p$$

$$p_{F,W} \leftarrow p_{F,W} + \delta p_F$$

# Contact Update: Covariance

- Use the Joseph-form covariance update:

$$P = (I - KH)P^-(I - KH)^T + KRK^T$$

- This is more numerically stable than:

$$P = (I - KH)P^-$$

- It helps preserve:

$$P = P^T, \quad P \succeq 0$$

- In code, symmetrize after the update:

$$P \leftarrow \frac{1}{2}(P + P^T)$$

# What Can Go Wrong?

- **IMU noise:** causes random drift after integration.
- **IMU bias:** causes systematic drift, often much worse than noise.
- **No contact:** the robot is temporarily dead-reckoning from IMU only.
- **Foot slip:** violates the fixed-contact assumption.
- **Wrong noise tuning:**
  - small  $Q$ : filter trusts prediction too much
  - large  $R$ : filter ignores foot updates
  - small  $R$ : filter over-trusts possibly slipping contacts

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# Tutorial Workflow

- 1 Run the provided simulation and inspect the data.
- 2 Visualize the EKF state and active foot states.
- 3 Implement IMU prediction.
- 4 Test IMU-only estimation.
- 5 Add IMU noise and bias.
- 6 Implement foot-contact correction.
- 7 Add foot slip and observe failure cases.
- 8 Tune  $Q$  and  $R$ .
- 9 Visualize covariance ellipses.
- 10 Apply the same estimator to real quadruped data.

## Task 1–2: Data and State Inspection

- Start with synthetic data. This provides ground truth for evaluation as follows:

$$e_p(t) = p_{\text{est}}(t) - p_{\text{gt}}(t)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{k=1}^N \|e_p(k)\|^2}$$

- Inspect the initialized state:

$$x = [R_{WB}, v_W, p_W]$$

- Then add stance-foot states using contact information:

$$x = [R_{WB}, v_W, p_W, p_{F,W}^{(1)}, \dots]$$

## Task 3: Implement IMU Propagation

- ① Propagate orientation:

$$R_k = R_{k-1} \exp([\omega_k]_{\times} \Delta t)$$

- ② Compute world-frame acceleration:

$$a_W = R_k a_B + g_W$$

- ③ Integrate velocity:

$$v_k = v_{k-1} + a_W \Delta t$$

- ④ Integrate position:

$$p_k = p_{k-1} + v_{k-1} \Delta t + \frac{1}{2} a_W \Delta t^2$$

- ⑤ Build  $F_k$  and propagate:

$$P_k^- = F_k P_{k-1} F_k^T + Q$$

## Task 4–5: IMU Noise and Bias

- Add random IMU noise (zero-mean Gaussian noise  $w_a, w_\omega$ ):

$$a_m = a + w_a, \quad \omega_m = \omega + w_\omega$$

- Add systematic IMU bias ( $b_a, b_\omega$ ):

$$a_m = a + b_a + w_a$$

$$\omega_m = \omega + b_\omega + w_\omega$$

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$$\omega_m = \omega + b_\omega + w_\omega$$

### Answers

- Noise causes drift because acceleration and angular velocity are integrated.
- Bias causes larger long-term drift due to accumulation over time.
- Gyro bias is especially harmful:

$$\omega_m = \omega + b_\omega \Rightarrow R_{WB} \text{ drifts over time} \Rightarrow a_W \text{ is rotated incorrectly} \Rightarrow p_W \text{ drifts}$$

## Task 6: Implement Foot-Contact Correction

- ① Compute predicted foot position:

$$\hat{p}_{F,W} = p_W + R_{WB}p_{F,B}$$

- ② Compute residual:

$$r = \hat{p}_{F,W} - p_{F,W}$$

- ③ Build measurement Jacobian:

$$H = [-R_{WB}[p_{F,B}]_{\times} \quad 0 \quad I \quad -I]$$

- ④ Compute:

$$S = HP^{-1}H^T + R, \quad K = P^{-1}H^TS^{-1}$$

- ⑤ Correct state:

$$\delta x = -Kr$$

- ⑥ Update covariance using Joseph form.

## Task 7: Foot Slip

- Introduce foot slip in the simulation.

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### Answer

- The contact update assumes:

$$p_{F,W} \approx \text{constant}$$

- Foot slip violates this assumption:

$$p_{F,W}(t + \Delta t) \neq p_{F,W}(t)$$

- Then the residual is no longer caused only by state-estimation error.
- The EKF may apply a wrong correction:

$$\delta x = -Kr$$

- Result:

- position error can increase,
- orientation can be pulled in the wrong direction,
- covariance can become overconfident.

## Task 8: Tuning $Q$ and $R$

- Process noise  $Q$  controls trust in prediction:

$$P^- = FPF^T + Q$$

- Measurement noise  $R$  controls trust in contact updates:

$$S = HP^-H^T + R$$

## Task 8: Tuning $Q$ and $R$

- Process noise  $Q$  controls trust in prediction:

$$P^- = FPF^T + Q$$

- Measurement noise  $R$  controls trust in contact updates:

$$S = HP^-H^T + R$$

### Answer

- If  $Q$  is too small:
  - the filter trusts the IMU prediction too much,
  - covariance becomes too small,
  - contact corrections have too little effect.
- If  $R$  is too large:
  - the filter ignores foot contacts,
  - behavior approaches IMU-only dead reckoning.
- If  $R$  is too small:

## Task 9: Covariance Ellipses

- The position covariance in the  $x$ - $y$  plane is:

$$P_{xy} = \begin{bmatrix} P_{xx} & P_{xy} \\ P_{yx} & P_{yy} \end{bmatrix}$$

- Covariance ellipses visualize uncertainty.

## Task 9: Covariance Ellipses

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- Covariance ellipses visualize uncertainty.

### Answer

- Ellipse size  $\sim$  uncertainty (larger = higher, smaller = lower)
- IMU-only prediction -  $P$  grows; Contact updates -  $P$  shrinks.
- During flight phases, no foot anchor is available, so uncertainty grows.

## Task 10: Real Data

- Apply the same EKF for real world data and tune the  $Q$  and  $R$  noise parameters.

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### Answer

- Real data is harder than simulation because of:
  - IMU bias and calibration errors,
  - time synchronization errors,
  - imperfect contact detection,
  - compliance and impacts,
  - kinematic errors,
  - foot slip,
  - uneven terrain.
- For rough terrain, possible improvements include:
  - contact probability estimation,
  - slip detection,
  - IMU bias estimation,
  - adaptive measurement noise,

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# Key Takeaways

- EKF combines:
  - IMU-based prediction (drift-prone),
  - contact-based corrections (drift-reducing).
- Stance feet act as temporary world-frame anchors.
- Performance depends strongly on:
  - noise tuning ( $Q$ ,  $R$ ),
  - contact quality (slip, detection),
  - modeling assumptions.
- Real-world data is more challenging than simulation.
- Error-state formulation enables efficient handling of orientation in  $SO(3)$ .